

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os
```

```
filename = '/content/78d8d060-21ce-11f0-89a9-cb5ed9bbefd9.jpg.webp'

if not os.path.exists(filename):
    raise ValueError(f"File '{filename}' does not exist.")

img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

if img is None:
    raise ValueError(f"Could not read '{filename}' as an image.")

rows, cols = img.shape
crow, ccol = rows // 2, cols // 2
```

```
dft = cv2.dft(np.float32(img), flags=cv2.DFT_COMPLEX_OUTPUT)
dft_shift = np.fft.fftshift(dft)

magnitude_spectrum = 20 * np.log(cv2.magnitude(dft_shift[:, :, 0], dft_shift[:, :, 1]) + 1)
```

```
radius_lp = 30
mask_lp = np.zeros((rows, cols, 2), np.uint8)
cv2.circle(mask_lp, (ccol, crow), radius_lp, (1, 1), -1)
```

```
array([[0, 0],
       [0, 0],
       [0, 0],
       ...,
       [0, 0],
       [0, 0],
       [0, 0]],

       [[0, 0],
       [0, 0],
       [0, 0],
       ...,
       [0, 0],
       [0, 0],
       [0, 0]],

       [[0, 0],
       [0, 0],
       [0, 0],
       ...,
       [0, 0],
       [0, 0],
       [0, 0]],

       ...,

       [[0, 0],
       [0, 0],
       [0, 0],
       ...,
       [0, 0],
       [0, 0],
       [0, 0]],

       [[0, 0],
       [0, 0],
       [0, 0],
       ...,
       [0, 0],
       [0, 0],
       [0, 0]],

       [[0, 0],
       [0, 0],
       [0, 0],
       ...,
       [0, 0],
       [0, 0],
       [0, 0]]], dtype=uint8)
```

```
fshift_lp = dft_shift * mask_lp
f_ishift_lp = np.fft.ifftshift(fshift_lp)
img_back_lp = cv2.idft(f_ishift_lp)
```

```
img_back_lp = cv2.magnitude(img_back_lp[:, :, 0], img_back_lp[:, :, 1])
img_back_lp = cv2.normalize(img_back_lp, None, 0, 255, cv2.NORM_MINMAX)
```

```
radius_hp = 30
mask_hp = np.ones((rows, cols, 2), np.uint8)
cv2.circle(mask_hp, (ccol, crow), radius_hp, (0, 0), -1)
```

```
array([[1, 1],
       [1, 1],
       [1, 1],
       ...,
       [1, 1],
       [1, 1],
       [1, 1]],

      [[1, 1],
       [1, 1],
       [1, 1],
       ...,
       [1, 1],
       [1, 1],
       [1, 1]],

      [[1, 1],
       [1, 1],
       [1, 1],
       ...,
       [1, 1],
       [1, 1],
       [1, 1]],

      ...,

      [[1, 1],
       [1, 1],
       [1, 1],
       ...,
       [1, 1],
       [1, 1],
       [1, 1]],

      [[1, 1],
       [1, 1],
       [1, 1],
       ...,
       [1, 1],
       [1, 1],
       [1, 1]],

      [[1, 1],
       [1, 1],
       [1, 1],
       ...,
       [1, 1],
       [1, 1],
       [1, 1]]], dtype=uint8)
```

```
fshift_hp = dft_shift * mask_hp
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img_back_hp = cv2.idft(f_ishift_hp)
img_back_hp = cv2.magnitude(img_back_hp[:, :, 0], img_back_hp[:, :, 1])
img_back_hp = cv2.normalize(img_back_hp, None, 0, 255, cv2.NORM_MINMAX)
```

```
plt.figure(figsize=(12, 8))
```

<Figure size 1200x800 with 0 Axes>
<Figure size 1200x800 with 0 Axes>

```
plt.subplot(2, 2, 1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1023.5), np.float64(575.5), np.float64(-0.5))
```

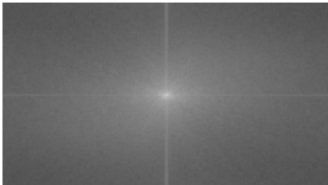
Original Image



```
plt.subplot(2, 2, 2)
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title('Magnitude Spectrum')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1023.5), np.float64(575.5), np.float64(-0.5))
```

Magnitude Spectrum



```
plt.subplot(2, 2, 3)
plt.imshow(img_back_lp, cmap='gray')
plt.title('Low-Pass Filtered Image')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1023.5), np.float64(575.5), np.float64(-0.5))
```

Low-Pass Filtered Image



```
plt.subplot(2, 2, 4)
plt.imshow(img_back_hp, cmap='gray')
plt.title('High-Pass Filtered Image')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1023.5), np.float64(575.5), np.float64(-0.5))
```

High-Pass Filtered Image



```
plt.tight_layout()
plt.savefig('output.png')
print("Saved output.png")
```

Saved output.png
<Figure size 640x480 with 0 Axes>

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